

Computational investigation of an α -hydroxyphosphonate derivative (DH4MPMP) for Anti-inflammatory and Autoimmune disease therapy

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Abstract



Inflammation has been prevalent worldwide for thousands of years, and advances in scientific research have revealed that chronic inflammation plays a key role in the development of numerous serious diseases, including autoimmune disorders. Current anti-inflammatory and autoimmune disease therapies are often associated with significant side effects, highlighting the need for safer and more effective therapeutic agents. Therefore, this study aimed to evaluate the potential of an α -hydroxyphosphonate derivative (DH4MPMP) as a candidate for anti-inflammatory and autoimmune disease therapy using an in silico molecular docking approach implemented in the Molecular Operating Environment (MOE) software. The compound's binding interactions were investigated against three key protein targets involved in inflammatory and autoimmune signaling pathways: JAK2, JAK3, and CXCR1. The docking results demonstrated promising binding affinities and interaction profiles compared with those of known inhibitors for these targets, suggesting that DH4MPMP may serve as a potential lead compound for the development of novel anti-inflammatory and autoimmune therapeutics. These encouraging findings warrant further experimental validation through in vitro and in vivo studies to confirm its efficacy and safety.

Molecular docking

To select therapeutic targets for our ligand, we used the SwissTarget software. The results obtained are illustrated in Figure 1. Among the targets given by the software we chose three targets which are JAK2, JAK3 and CXCR1 to test the anti-inflammatory effect of our synthesized molecule.

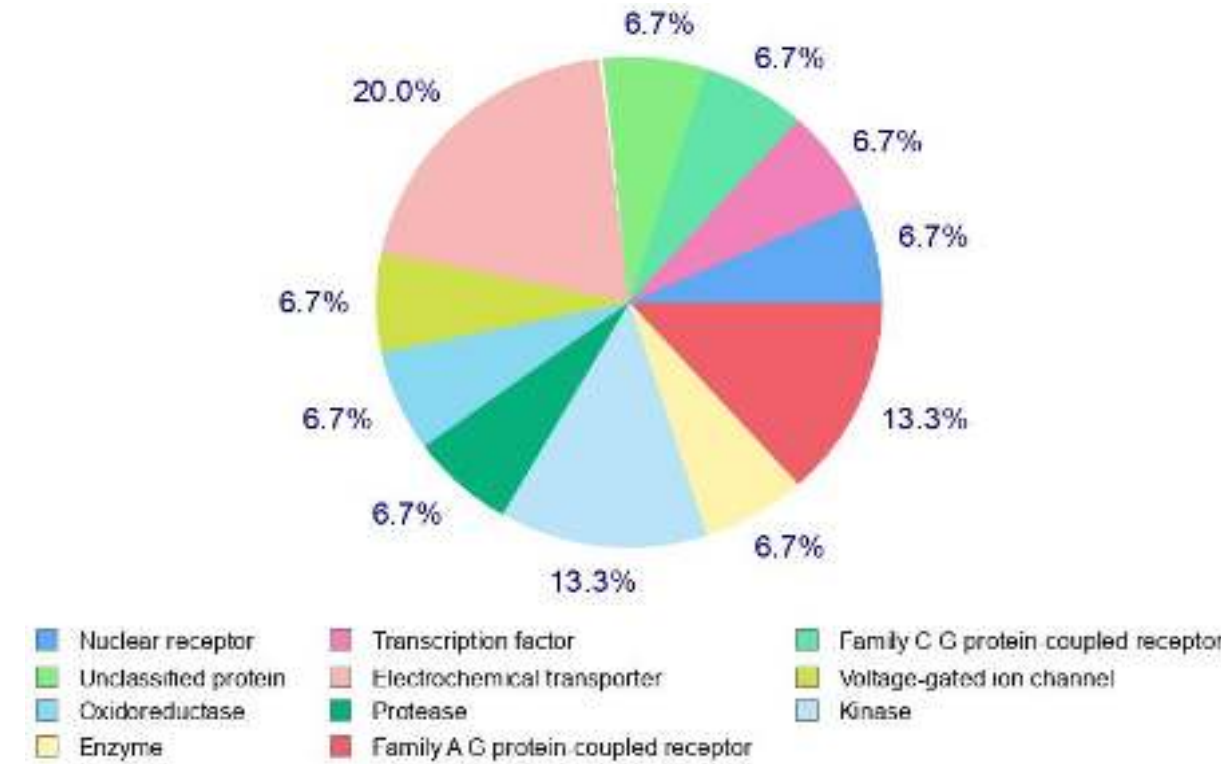


Figure 1 : Target Selection.

Results and discussions

Docking of DH4MPMP and native ligand with JAK2

During the docking calculations, we determined 10 poses. Here we present the best pose (Figure 2) which corresponds to the minimum energies with an RMSD of 1.41 Å of the ligand in the JAK2 protein (Table1). Figure 2 shows the 2D interactions between the DH4MPMP ligand and the JAK2 protein; note the presence of H-acceptor interactions with several residues such as Asn673, Asn678, and Thr555.

The interaction between JAK2 and JH2 exhibits spatial proximity at the binding site. The JH2 inhibitor forms an H-acceptor bond with the amino acid tyrosine, where the O group of JH2 is close to the tyrosine. Furthermore, another interaction occurs between the O group of JH2 and lysine, where the bond is also an H-acceptor. These interactions are crucial for stabilizing the JAK2-JH2 interaction.

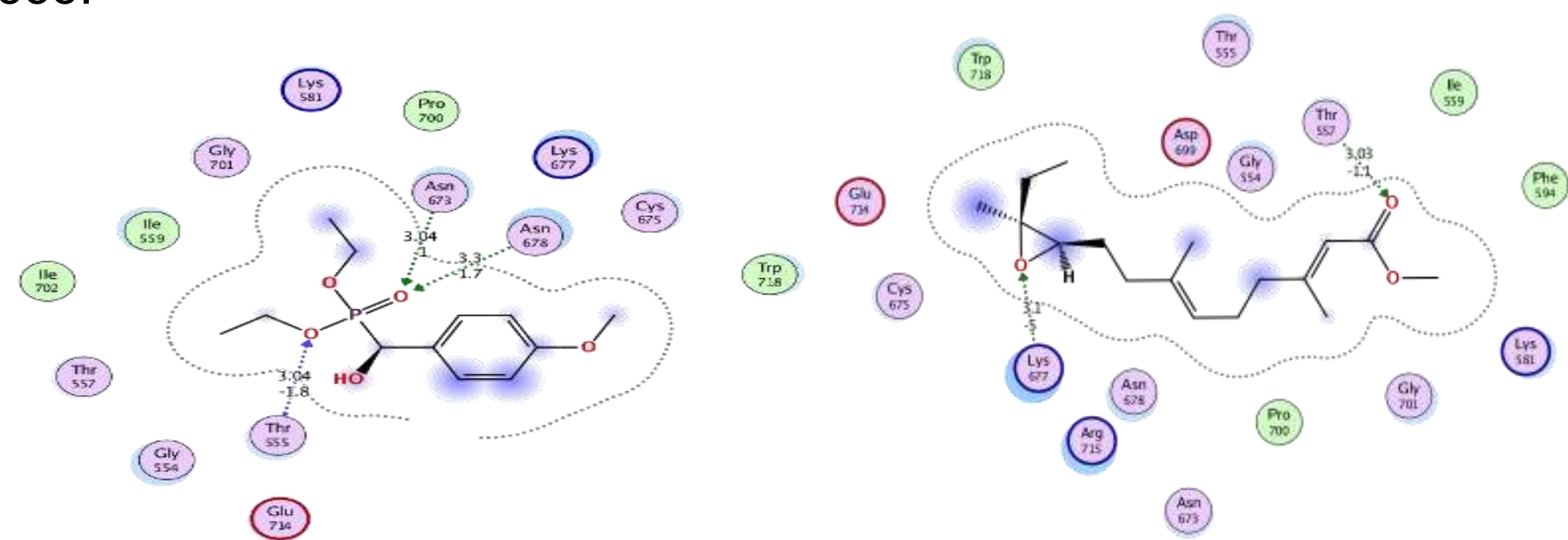


Figure 2. Best interaction position of JAK3 with DH4MPMP and Leflunomide

Docking of DH4MPMP and native ligand with JAK2

Figure 3 shows the best pose resulting from the docking of our ligand (DH4MPMP) with the JAK3 protein. This pose corresponds to minimum energies with an RMSD of 1.20 Å (Table1). Three types of interactions were identified: the H- π interaction with the Phe833 residue, the H-Donor interaction with the Glu871 residue, and the H-Acceptor interaction with the Lys855 residue.

The inhibitor leflunomide forms a donor H-type bond with the amino acid Asp967, where the N6 group of leflunomide is located near this residue. Furthermore, another interaction of the same type occurs between the C28 group of leflunomide and the same residue. The leflunomide ring exhibits a π -cation-type interaction with the Lys855 residue (Figure 3).

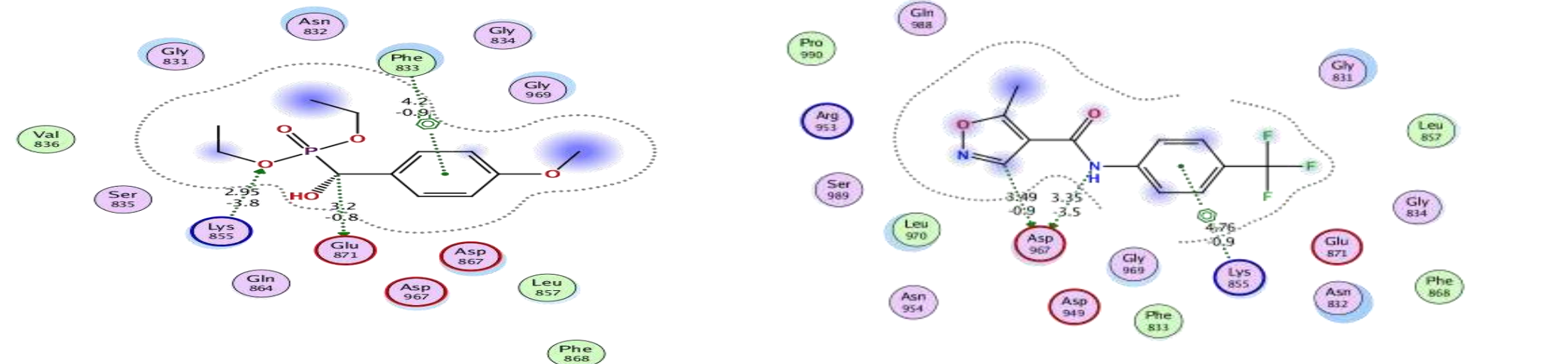


Figure 3. Best interaction position of JAK3 with DH4MPMP and Leflunomide.

Docking of DH4MPMP and native ligand with CXCR1

The best pose obtained after docking our ligand (DH4MPMP) with the CXCR1 protein is illustrated in Figure III.8. We selected this pose based on its RMSD, which is 1.10 Å (Table 1). Two types of interactions were identified: the H- π interaction with residue phe211 and the π - π interaction with residue phe251 (Figure 4).

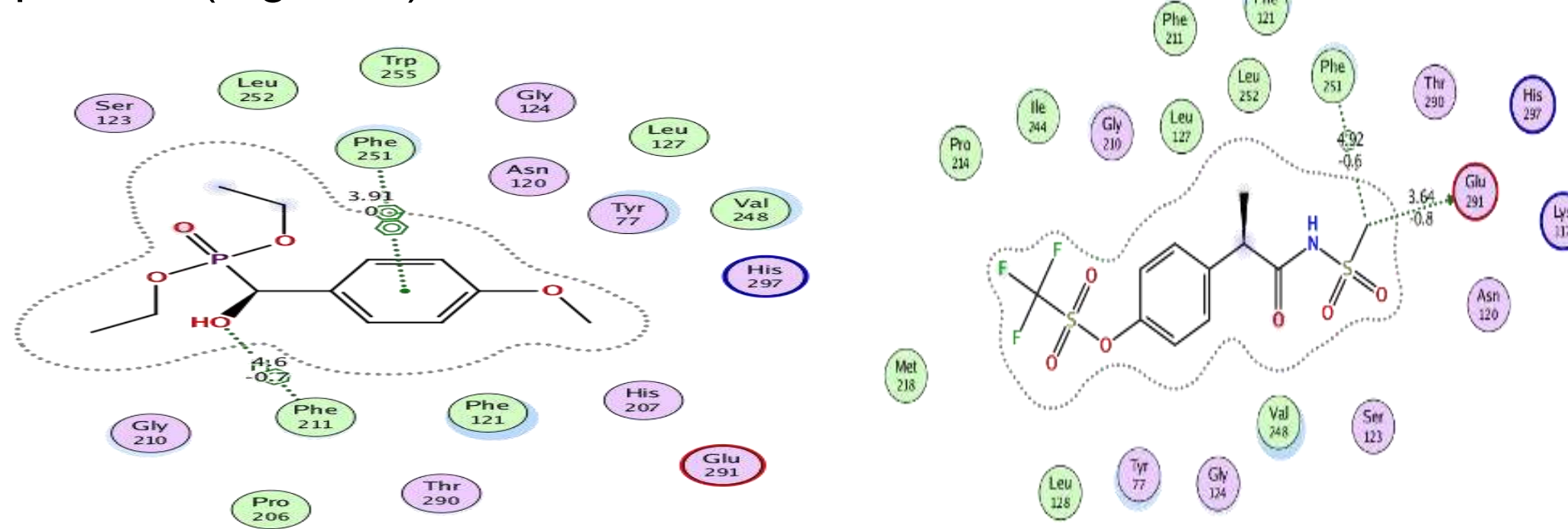


Figure 4. Best interaction position of CXCR1 with DH4MPMP and Leflunomide.

Table 1. Binding energy, distance and molecular interactions of the two ligands with three proteins.

Protein	Ligand	Score (Kcal/mol)	RMSD	Interactions		
				Atom	Residues	Type of liaison
JAK2	DH4MPMP	-5.74	1.41	O ₁₈	Asn673	H-Acceptor
				O ₁₈	Asn678	H-Acceptor
				O ₃₆	Thr555	H-Acceptor
				O ₄	Thr557	H-Acceptor
				O ₆	Lys677	H-Acceptor
JAK3	DH4MPMP	-6.25	1.20	Cycle	Phe833	H- π
				C ₁₁	Glu871	H-Donnor
				O ₃₆	Lys855	H-Acceptor
	Leflunomide	-5.20	1.55	Cycle	Lys855	π -Cation
				N ₆	Asp967	H-Donnor
				C ₁₈	Asp967	H-Donnor
CXCR1	DH4MPMP	-6,50	1.10	O ₃₄	Phe211	H- π
				Cycle	Phe251	π - π
	Ladarixine	-6,67	1.76	C ₂₂	Phe251	H- π
				C ₂₂	Glu291	H-Acceptor

Table 2 presents a comparison of the ADMET results for the ligand DH4MPMP and the JAK2 protein inhibitor (JH2). DH4MPMP and JH2 exhibit human intestinal absorption with probabilities of 0.94 and 0.99, respectively. Therefore, both the ligand and the JAK2 inhibitor have a high probability of intestinal absorption. Both inhibitors have similar probabilities of crossing the blood-brain barrier. They do not exhibit respiratory, hepatic, or renal toxicity.

Table 2. Comparison of ADMET results for DH4MPMP and the JAK2 inhibitor (JH2)

Classification	C ₁₂ H ₁₉ O ₅ P		C ₁₇ H ₂₈ O ₅	
	Value	Probability	Value	Probability
Human intestinal absorption	+	0.9393	+	0.9893
Blood-brain barrier	+	0.7250	+	0.8000
Carcinogenicity (binary)	-	0.7428	-	0.7100
Eye corrosion	-	0.8129	-	0.9495
Hepatotoxicity	-	0.6734	-	0.7000
Skin sensitization	-	0.8515	+	0.5842
Respiratory toxicity	-	0.5889	-	0.6222
Nephrotoxicity	-	0.5315	-	0.4384
Estrogen receptor binding	+	0.5771	-	0.5430
Androgen receptor binding	+	0.5426	+	0.5525
Thyroid receptor binding	-	0.5562	-	0.5187
Biodegradation	-	0.7500	-	0.7500

Table 3 presents a comparison of the ADMET results for the ligand DH4MPMP and the JAK3 inhibitor (Leflunomide). DH4MPMP and Leflunomide exhibit human intestinal absorption with probabilities of 0.94 and 0.99, respectively. Therefore, we can conclude that both the ligand and the JAK2 inhibitor have a high probability of intestinal absorption. Both inhibitors have similar probabilities of crossing the blood-brain barrier. They do not exhibit respiratory, hepatic, or renal toxicity.

Table 3 Comparison of ADMET results for DH4MPMP and the JAK3 inhibitor (Leflunomide)

Classification	C ₁₂ H ₁₉ O ₅ P		C ₁₇ H ₂₈ O ₅	
	Value	Probability	Value	Probability
Human intestinal absorption	+	0.9393	+	0.9907
Blood-brain barrier	+	0.7250	+	0.9000
Carcinogenicity (binary)	-	0.7428	-	0.6219
Eye corrosion	-	0.8129	-	0.9886
Hepatotoxicity	-	0.6734	-	0.9000
Skin sensitization	-	0.8515	-	0.8627
Respiratory toxicity	-	0.5889	+	0.7000
Nephrotoxicity	-	0.5315	-	0.2205
Estrogen receptor binding	+	0.5771	+	0.8905
Androgen receptor binding	+	0.5426	+	0.6823
Thyroid receptor binding	-	0.5562	+	0.7701
Biodegradation	-	0.7500	-	0.8250

Table 4 presents a comparison of the ADMET results for the ligand DH4MPMP and the CXCR1 protein inhibitor (ladarixin). DH4MPMP and ladarixin exhibit human intestinal absorption with probabilities of 0.94 and 0.97, respectively. Therefore, we can conclude that both the ligand and the JAK2 inhibitor have a high probability of intestinal absorption. Both inhibitors have similar probabilities of crossing the blood-brain barrier. They do not exhibit respiratory, hepatic, or renal toxicity.

Table 4 Comparison of ADMET results for DH4MPMP and the CXCR1 inhibitor (ladarixin).

Classification	C ₁₂ H ₁₉ O ₅ P		C ₁₁ H ₁₂ F ₃ NO ₆ S ₂	
	Value	Probability	Value	Probability
Human intestinal absorption	+	0.9393	+	0.9724
Blood-brain barrier	+	0.7250	+	0.7250
Carcinogenicity (binary)	-	0.7428	-	0.7050
Eye corrosion	-	0.8129	-	0.9018
Hepatotoxicity	-	0.6734	-	0.1000
Skin sensitization	-	0.8515	-	0.8632
Respiratory toxicity	-	0.5889	+	0.6778
Nephrotoxicity	+	0.4685	+	0.5503
Estrogen receptor binding	+	0.5771	+	0.6716
Androgen receptor binding	+	0.5426	+	0.6394
Thyroid receptor binding	-	0.5562	-	0.7031
Biodegradation	-	0.7500	-	0.8250

Conclusion

The results of molecular docking of the ligand alone and with the protein inhibitors used showed us the interactions of this ligand with different residues and with several types of binding. Using Lipinski's rule as a reference, we can say that the DH4MPMP ligand could be used as a drug compared to the three protein inhibitors already on the market. The ADMET evaluation results indicate that the DH4MPMP ligand exhibits human intestinal absorption with a probability of 0.94, indicating good absorption through the intestinal wall. It has the ability to cross the blood-brain barrier with a probability of 0.72, meaning it can enter the brain from the bloodstream. It does not exhibit respiratory, hepatic, or renal toxicity. The information obtained on the molecular interactions between Diethyl [Hydroxy (4-MethoxyPhenyl) Methyl] Phosphonate and the proteins JAK2, JAK3 and CXCR1 opens up perspectives for further research and may eventually lead to the development of innovative treatments for inflammatory diseases.

References

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